



**CALICUT UNIVERSITY – FOUR-YEAR UNDER  
GRADUATE PROGRAMME (CU-FYUGP)**

**BSc PHYSICS HONOURS**

|                |                                                                                                                                                                                                      |                     |                      |                       |                |
|----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|----------------------|-----------------------|----------------|
| Programme      | <b>B.Sc. Physics Honours</b>                                                                                                                                                                         |                     |                      |                       |                |
| Course Title   | <b>MATHEMATICAL METHODS FOR PHYSICS</b>                                                                                                                                                              |                     |                      |                       |                |
| Type of Course | <b>Minor (GROUP I: MATHEMATICS FOR PHYSICAL SYSTEMS)</b>                                                                                                                                             |                     |                      |                       |                |
| Semester       | <b>III</b>                                                                                                                                                                                           |                     |                      |                       |                |
| Academic Level | <b>200 –299</b>                                                                                                                                                                                      |                     |                      |                       |                |
| Course Details | Credit                                                                                                                                                                                               | Lecture<br>per week | Tutorial<br>per week | Practical<br>per week | Total<br>Hours |
|                | 4                                                                                                                                                                                                    | 3                   | -                    | 2                     | 75             |
| Pre-requisites | Fundamentals of vectors, linear algebra, differential equations coordinate systems and familiarity with basic concepts in physics.                                                                   |                     |                      |                       |                |
| Course Summary | This course explores fundamental principles and applications of vector analysis, complex functions, differential equations and curvilinear coordinates in electromagnetism and engineering contexts. |                     |                      |                       |                |

**Course Outcomes (CO):**

| CO  | CO Statement                                                                                                                                        | Cognitive Level* | Knowledge Category# | Evaluation Tools used           |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|---------------------------------|
| CO1 | Students will attain a strong foundational understanding about vector calculus, complex numbers, differential equations and curvilinear coordinates | U                | C                   | Instructor-created exams / Quiz |

|                                                                                                                                                                                                           |                                                                                                                                                                         |    |       |                                                                           |
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| CO2                                                                                                                                                                                                       | Students will develop analytical proficiency which enables them to analyse and interpret complex physical phenomena through the application of mathematical principles. | Ap | P & M | Practical Assignment / Observation of Practical Skills                    |
| CO3                                                                                                                                                                                                       | Students will cultivate advanced problem-solving skills.                                                                                                                | Ap | P     | Practical Assignment / Observation of Practical Skills                    |
| CO4                                                                                                                                                                                                       | Students will enhance their ability to model and represent physical systems mathematically for describing and understanding complex phenomena.                          | Ap | P M   | Practical Assignment / Observation of Practical Skills / Home Assignments |
| CO5                                                                                                                                                                                                       | Students will recognize and appreciate the interdisciplinary applications of mathematical methods.                                                                      | Ap | C & M | Seminar Presentation / Group Discussion                                   |
| CO6                                                                                                                                                                                                       | Students will refine their critical thinking which encourages independent inquiry and problem-solving approaches in tackling challenging problems and scenarios.        | Ap | P & M | Group Discussion/ Viva Voce                                               |
| <p>* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)</p> <p># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)</p> |                                                                                                                                                                         |    |       |                                                                           |

**Detailed Syllabus:**

| <b>Module</b> | <b>Unit</b>                                  | <b>Content</b>                                                                                                                                               | <b>Hrs<br/>(45<br/>+30<br/>)</b> | <b>Marks<br/>(70)</b> |
|---------------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|-----------------------|
| <b>I</b>      | <b>VECTOR CALCULUS</b>                       |                                                                                                                                                              | <b>12</b>                        | <b>20</b>             |
|               | 1                                            | Scalar and Vector Point Functions, Gradient of a Scalar Function<br>Geometrical Meaning of Gradient                                                          | 4                                |                       |
|               | 2                                            | Normal and Directional Derivative, Divergence of a Vector Function,<br>Physical Interpretation of Divergence, Divergence and Curl of<br>Electrostatic Fields | 4                                |                       |
|               | 3                                            | Curl, Physical Meaning of Curl, The Divergence and Curl of B                                                                                                 | 4                                |                       |
|               |                                              | Sections 2.4, to 2.11 of book 2, Sections 2.2.1 – 2.2.4 of chapter 2 and<br>Section 5.3.1 – 5.3.3 of chapter 5 of book 1                                     |                                  |                       |
| <b>II</b>     | <b>COMPLEX NUMBERS AND COMPLEX FUNCTIONS</b> |                                                                                                                                                              | <b>11</b>                        | <b>15</b>             |
|               | 4                                            | Introduction, Complex Numbers                                                                                                                                | 1                                |                       |
|               | 5                                            | Geometrical Representation of Imaginary Numbers Argand Diagram                                                                                               | 1                                |                       |
|               | 6                                            | Equal Complex Numbers, Addition, Addition of Complex Numbers<br>by Geometry                                                                                  | 1                                |                       |
|               | 7                                            | Subtraction, Powers of $i$ , Multiplication, $i$ (Iota) as an<br>Operator, Conjugate of a Complex Number                                                     | 1                                |                       |
|               | 8                                            | Division, Division of Complex numbers by Geometry                                                                                                            | 1                                |                       |
|               | 9                                            | Modulus and Argument, Polar form, Types of Complex Numbers                                                                                                   | 1                                |                       |
|               | 10                                           | Resistance and Reactance                                                                                                                                     | 2                                |                       |
|               | 11                                           | The L-R-C series Circuit                                                                                                                                     | 3                                |                       |
|               |                                              | Sections 20.1 to 20.17 of book2, Sections 31.2 and 31.3 of book 3                                                                                            |                                  |                       |
| <b>III</b>    | <b>ORDINARY DIFFERENTIAL EQUATIONS</b>       |                                                                                                                                                              | <b>12</b>                        | <b>20</b>             |
|               | 12                                           | Definition, order and Degree of a Differential Equation                                                                                                      | 1                                |                       |
|               | 13                                           | Formation of Differential Equations, Solution of a Differential<br>Equation                                                                                  | 1                                |                       |

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|    | 14                                                                                         | Geometrical Meaning of the Differential Equation of the First order and First Degree, Differential Equations of the First order and First Degree                                                                                                                                                                  | 2         |           |
|    | 15                                                                                         | Variables Separable, Homogeneous Differential Equations, Equations Reducible to Homogeneous form, Linear Differential Equations, Equations Reducible to the Linear form (Bernoulli Equation)                                                                                                                      | 4         |           |
|    | 16                                                                                         | Non-Linear Differential Equations, Linear Differential Equations of Second order with Constant Coefficients                                                                                                                                                                                                       | 2         |           |
|    | 17                                                                                         | Periodic Motion- Simple Harmonic motion. Applications of simple Harmonic motion, Damped oscillations                                                                                                                                                                                                              | 2         |           |
|    | Sections 12.1 to 12.11, 13.2, 13.3 of book 2, Sections 14.2, 14.4, 14.7 of Book 3          |                                                                                                                                                                                                                                                                                                                   |           |           |
| IV | <b>CURVILINEAR COORDINATES</b>                                                             |                                                                                                                                                                                                                                                                                                                   | <b>10</b> | <b>15</b> |
|    | 18                                                                                         | Curvilinear Coordinates                                                                                                                                                                                                                                                                                           | 1         |           |
|    | 19                                                                                         | Cylindrical (Polar) Co-ordinates                                                                                                                                                                                                                                                                                  | 2         |           |
|    | 20                                                                                         | Spherical Polar Co-ordinates                                                                                                                                                                                                                                                                                      | 2         |           |
|    | 21                                                                                         | Relation Between Cylindrical and Spherical Co-ordinates                                                                                                                                                                                                                                                           | 2         |           |
|    | 22                                                                                         | Applications of Gauss's Law in polar, cylindrical and spherical problems                                                                                                                                                                                                                                          | 3         |           |
|    | Sections 4.1, 4.8, 4.9, 4.12 of book 2, Section 2.2.3 Application of Gauss's law of Book 1 |                                                                                                                                                                                                                                                                                                                   |           |           |
| V  | <b>PRACTICALS</b>                                                                          |                                                                                                                                                                                                                                                                                                                   | <b>30</b> |           |
|    | 1                                                                                          | <b>Flywheel- Determination of the Moment of Inertia.</b> <ul style="list-style-type: none"> <li>This experiment aims to help students grasp the concept of energy conservation and the dynamics of rotation.</li> <li>Do at least 9 trials for different masses and number of turns wound on the axil.</li> </ul> |           |           |
|    | 2                                                                                          | <b>Torsion Pendulum- Determination of the Moment of Inertia.</b> <ul style="list-style-type: none"> <li>Using identical masses on the disc, determine the moment of inertia of the disc.</li> <li>Verify the moment of inertia by direct method, <math>I = \frac{1}{2}MR^2</math></li> </ul>                      |           |           |

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| 3 | <b>Compound Pendulum- Acceleration Due to Gravity and Moment of Inertia and Verification of Parallel Axis Theorem.</b> <ul style="list-style-type: none"> <li>Plot a graph of distance of knife edge from one end Vs period of oscillations. Using the measurement from the graph, calculate g.</li> <li>Calculate the radius of gyration and hence the moment of inertia about CM. Compare the result obtained by the direct calculation <math>I_{CM} = \frac{ML^2}{12}</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                   |  |  |
| 4 | <b>Kater's Pendulum- Determination of Earth's Gravity.</b> <ul style="list-style-type: none"> <li>To determine g and discuss the relative merits of both cases by estimation of error in the two cases.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |  |  |
| 5 | <b>Sonometer - Determine the Frequency of AC.</b> <ul style="list-style-type: none"> <li>Estimate the linear mass density of the wire.</li> <li>Draw <math>L^2 - m</math> graph and from the slope calculate the frequency.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |
| 6 | <b>Determination of the Velocity of Sound in Air.</b> <ul style="list-style-type: none"> <li>Sound wave of known frequency is generated using a wave generator(WG) and piezo buzzer and are recorded using a microphone(MIC).</li> <li>Phase differences between the WG and MIC waveforms were analyzed in a CRO and the distance between them were adjusted to make both of them in phase and hence calculate velocity of sound.</li> <li>Phase difference can be analyzed from the Lissajous figure obtained by X-Y plotting of WG and MIC waves.</li> <li>ExpEYES may be used.</li> <li><a href="https://expeyes.in/experiments/sound/velocity.html">https://expeyes.in/experiments/sound/velocity.html</a></li> <li><a href="https://expeyes.in/experiments/electrical/xyplot.html">https://expeyes.in/experiments/electrical/xyplot.html</a></li> </ul> |  |  |

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| 7 | <p><b>Pendulum- Limits on Angular Displacement and Study of Damped Oscillations.</b></p> <ul style="list-style-type: none"> <li>• Estimate limits on angular displacement for SHM by measuring the time period at different angular displacements and compare it with the expected value of time period for SHM.</li> <li>• Study damped oscillations. Plot amplitude as a function of time and determine the damping coefficient and Q factor.</li> <li>• Digitized data can be used for the study.</li> <li>• <a href="https://www.youtube.com/watch?v=jcpvm95bhXw">https://www.youtube.com/watch?v=jcpvm95bhXw</a></li> <li>• <a href="https://expeyes.in/experiments/school-level/sr04.html">https://expeyes.in/experiments/school-level/sr04.html</a></li> <li>• <a href="https://phyphox.org/experiment/pendulum/">https://phyphox.org/experiment/pendulum/</a></li> </ul> |  |  |
| 8 | <p><b>Black body spectrum of Sun -Estimation of surface temperature using the Tracker Video Analysis tool.</b></p> <ul style="list-style-type: none"> <li>• Calibrate the video of the solar spectra in the Tracker tool using two laser wavelengths/lines of mercury spectra.</li> <li>• Plot wavelength vs intensity, get <math>\lambda_{max}</math> and using Wein's law calculate the surface temperature.</li> <li>• Pre recorded video of the solar spectra can be used.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                        |  |  |
| 9 | <p><b>Analysis of Hydrogen spectra using the Tracker Video Analysis tool.</b></p> <ul style="list-style-type: none"> <li>• Calibrate the video of the Hydrogen spectra in the Tracker tool using two laser wavelengths/lines of mercury spectra.</li> <li>• Plot the intensity profile, find the prominent wavelengths of the Balmer series and calculate the Rydberg's constant.</li> <li>• Estimate the %error.</li> <li>• Pre recorded video of the Hydrogen spectra can be used.</li> <li>• <a href="https://physlets.org/tracker/">https://physlets.org/tracker/</a>.</li> <li>• <a href="https://www.youtube.com/watch?v=UCCPkIpUQEw">https://www.youtube.com/watch?v=UCCPkIpUQEw</a></li> </ul>                                                                                                                                                                           |  |  |

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|  | 10 | <b>RC and RL transients - determination of capacitance and inductance.</b> <ul style="list-style-type: none"> <li>• Apply a voltage step to a series RC/RL circuit and record the resulting voltage variation across the capacitor/inductor.</li> <li>• Get the value of time constant by an exponential fit to the data.</li> <li>• Repeat the experiment for different resistances.</li> <li>• <a href="https://expeyes.in/experiments/electrical/rctransient.html">https://expeyes.in/experiments/electrical/rctransient.html</a></li> <li>• <a href="https://expeyes.in/experiments/electrical/rltransient.html">https://expeyes.in/experiments/electrical/rltransient.html</a></li> </ul> |  |  |
|  | 11 | <b>Determination of Plank's constant using LEDs</b> <ul style="list-style-type: none"> <li>• Observe the turn-on voltage,</li> <li>• <math>V_0</math> of LEDs and calculate the value of <math>h</math>. Use at least 4 different colors of LED (with transparent casing)</li> <li>• Plot <math>\frac{1}{\lambda} - V_0</math> graph using Python, fit a straight line to get the slope and estimate the value of <math>h</math>.</li> <li>• Calculate the %error.</li> <li>• Programmable voltage source of ExpEYES may be used to find the turn-on voltage.</li> </ul>                                                                                                                       |  |  |
|  | 12 | <b>Construction of the center tapped full wave rectifiers and regulated power supply</b> <ul style="list-style-type: none"> <li>• Construct a center tapped full wave rectifier without filter and with a filter.</li> <li>• Measure the AC and DC voltages using a multimeter and calculate the ripple factor without and with a filter.</li> <li>• Observe the variation of the ripple factor with load resistance, when filter is used.</li> <li>• Construct 5V/12V regulated power supply using 78XX IC.</li> </ul>                                                                                                                                                                        |  |  |
|  | 13 | <b>Construct Half adder using universal gates and study the operation.</b> <ul style="list-style-type: none"> <li>• Implement half adder using NAND/NOR gates and verify the truth table for each input/output combination.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                         |  |  |

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| 14 | <b>Verification of De-Morgan's Theorems using basic gates.</b> <ul style="list-style-type: none"> <li>Realize the either side of the De-Morgan's Theorems using gates from appropriate ICs and verify the truth table for each input/output combination.</li> </ul>                                                                                                                                                                                                                                              |  |  |
| 15 | <b>Construction of the center tapped full wave rectifiers and regulated power supply.</b> <ul style="list-style-type: none"> <li>Construct a center tapped full wave rectifier without filter and with a filter.</li> <li>Measure the AC and DC voltages using a multimeter and calculate the ripple factor without and with a filter.</li> <li>Observe the variation of the ripple factor with load resistance, when filter is used.</li> <li>Construct 5V/12V regulated power supply using 78XX IC.</li> </ul> |  |  |

**Books and References:**

1. Introduction to Electrodynamics by David J Griffiths, 5<sup>th</sup> Edition (Book 1)
2. Mathematical Physics by H K Das and Rama Verma, 7<sup>th</sup> Edition (Book 2)
3. University Physics With Modern Physics by Hugh D Young and Roger A Freedman 14<sup>th</sup> edition (Book 3)
4. Mathematical Physics by Satya Prakash - S Chand and Sons

**Mapping of COs with PSOs and POs:**

|      | PS<br>O1 | PS<br>O2 | PS<br>O3 | PSO<br>4 | PS<br>O5 | PS<br>O6 | PO<br>1 | PO<br>2 | PO<br>3 | PO<br>4 | PO<br>5 | PO<br>6 | PO<br>7 |
|------|----------|----------|----------|----------|----------|----------|---------|---------|---------|---------|---------|---------|---------|
| CO 1 | 3        | 1        | 2        | 0        | 2        | 2        | 3       | 2       | 2       | 1       | 3       | 2       | 1       |
| CO 2 | 2        | 3        | 2        | 1        | 1        | 1        | 2       | 2       | 2       | 1       | 3       | 2       | 0       |
| CO 3 | 1        | 2        | 3        | 1        | 2        | 1        | 2       | 2       | 2       | 1       | 3       | 2       | 1       |
| CO 4 | 2        | 1        | 1        | 3        | 2        | 1        | 2       | 2       | 2       | 1       | 3       | 2       | 0       |
| CO 5 | 2        | 2        | 2        | 1        | 3        | 1        | 2       | 2       | 2       | 1       | 3       | 2       | 1       |
| CO 6 | 2        | 1        | 3        | 0        | 2        | 3        | 2       | 2       | 2       | 1       | 3       | 2       | 1       |

**Correlation Levels:**

| Level | Correlation        |
|-------|--------------------|
| 0     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Quiz / Discussion / Seminar
- Internal Theory/Practical Exam
- Assignments /Viva
- End Semester Exam (70%)

**Mapping of COs to Assessment Rubrics**

|      | Internal Theory/<br>Practical Exam | Assignment /Viva | Practical Skill<br>Evaluation | End Semester<br>Examinations |
|------|------------------------------------|------------------|-------------------------------|------------------------------|
| CO 1 | ✓                                  | ✓                |                               | ✓                            |
| CO 2 | ✓                                  | ✓                |                               | ✓                            |
| CO 3 | ✓                                  | ✓                |                               | ✓                            |
| CO 4 | ✓                                  | ✓                |                               | ✓                            |
| CO 5 | ✓                                  | ✓                |                               | ✓                            |
| CO 6 |                                    | ✓                | ✓                             |                              |